

GIS Based Road Transportation Network Analysis



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Network Analysis: A definition

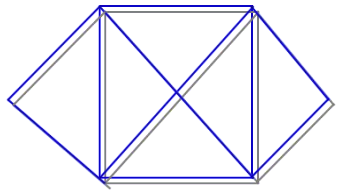
Any system of interconnected features through which the resources flow/ communication achieved. Representing some geographic phenomenon.

- A graph is a defined as G= (V, E)
- V(G) = {V1, V2, …, Vn}, E(G) = {(Vi, Vj)} or E(G) = {E1, E2, …., Em}

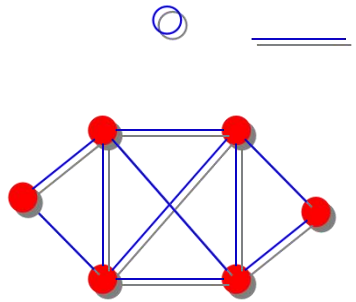
Features: Only point & Line, No polygon Feature

Resource: Anything which is moving through these features

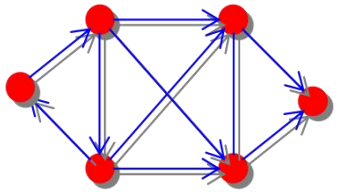
Also called linear complex, 1D-complex



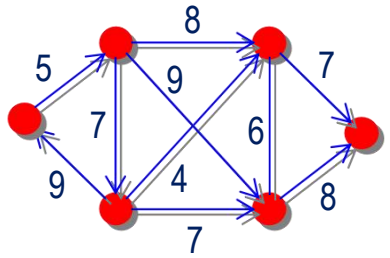
Connected, Un-weighted
Planar, Undirected



Disconnected, Un-weighted
Planar, Undirected



Connected, Un-weighted
Planar, Directed



Connected, Weighted
Planar, Directed

- Connected vs. Disconnected Graphs
- Directed vs. Undirected Graphs
- Planar vs. Non-planar Graphs
- Weighted vs. Un-weighted Graphs

Similarity with GIS:
In GIS point and line features can be created very easily and analysis can be performed

Weight Function or Cost Function Cost Factor or Impedance Factor

Type: Unidirectional/Bidirectional, Positive/Negative , From-To & To-From

Unit: Unit of Time or Distance

Network Related Problems/Applications

1. Shortest Path Problem
   - *Driving Time Analysis*
   - *Optimal Routing*
   - *Closest facility analysis*
   - *Service Area Analysis*
   - *Driving Directions*
2. Supply-Demand Problem (TSP/VRP)
3. Minimum Spanning Tree (MST) Problem
4. Chinese Postman Problem (CPP)
5. Maximum Flow Problem (MFP)



1. Shortest Path Problem

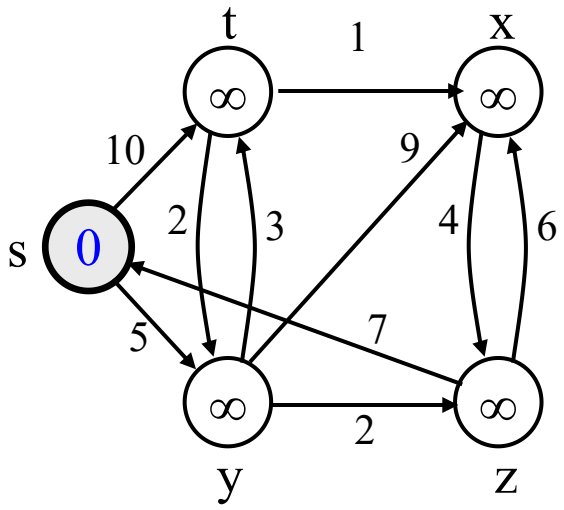
Path: A path in a graph G is a sequence of nodes $x_1, x_2, \ldots, x_k$, such that there is an edge from each node the next one in the sequence.

Shortest Path: A path of minimum impedance (minimum cost).

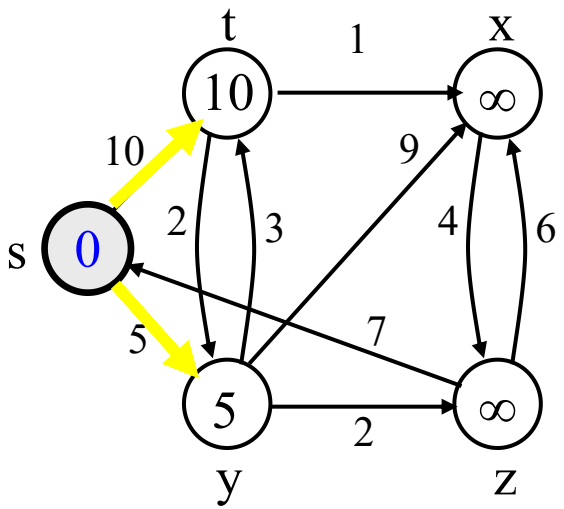
Dijkstra's Algorithm

Example

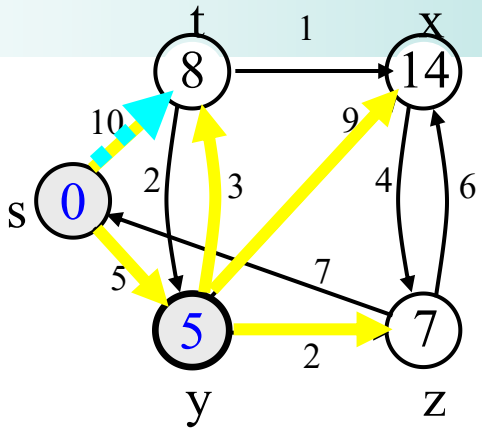
S=<> Q=<s,t,x,z,y>



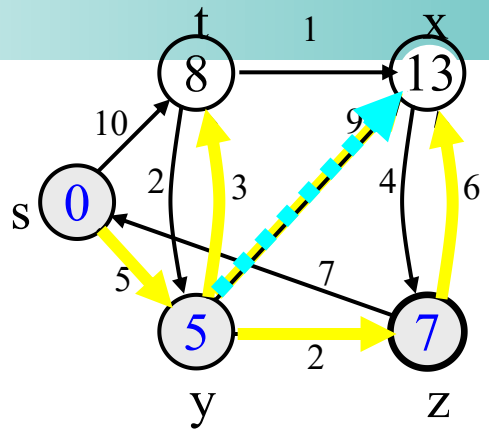
S=<s> Q=<y,t,x,z>



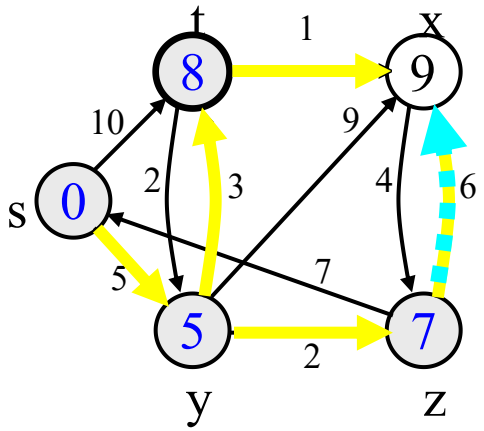
Example (cont.)



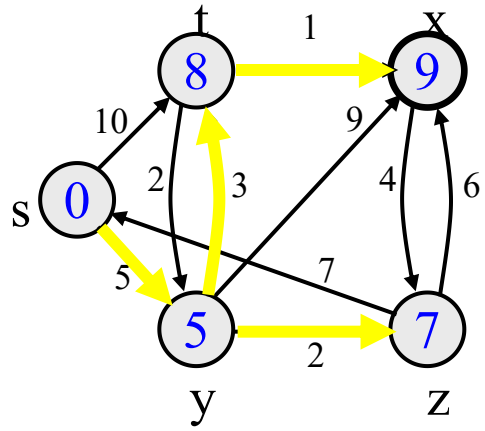
S = <s,y> Q = <z,t,x>



S = <s,y,z> Q=<t,x>



S = <s,y,z,t> Q = <x>

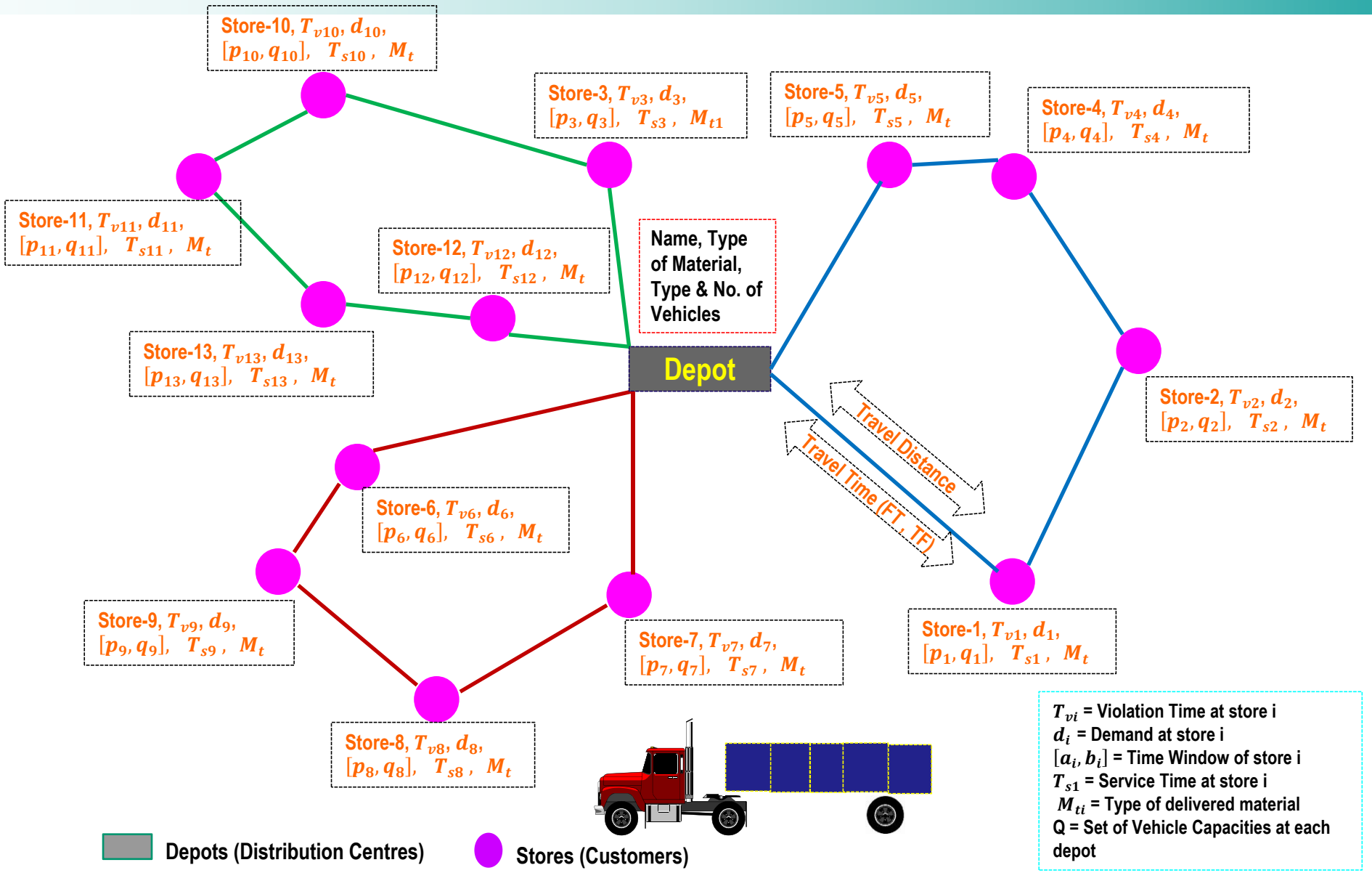


S = <s,y,z,t,x> Q=<>

Successive Paths?

1) From a given (fixed) Origin (O) to a given Destination (D)
2) From a given Origin (O) to any Destination (D)
3) From any Origin (O) to a given Destination (D)
4) From any Origin (O) to any Destination (D) .....Floyd's Algorithm

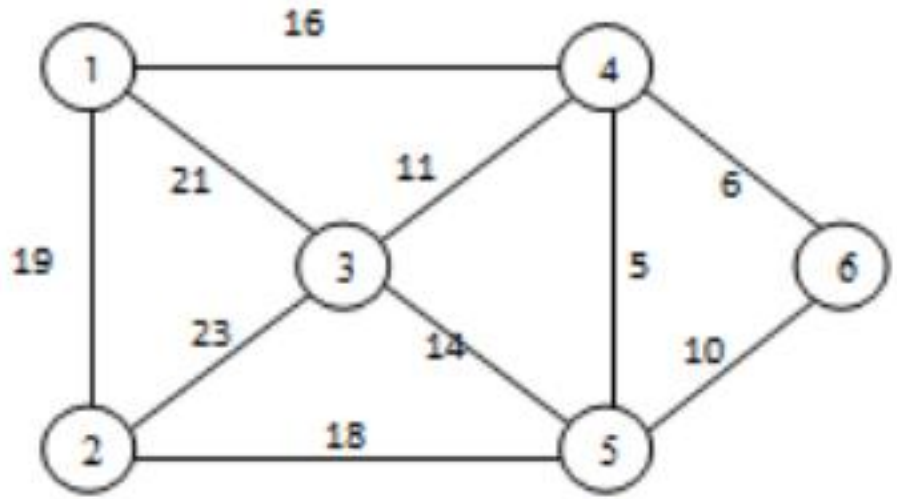
2. Supply-Demand Problem: Vehicle Routing Problem



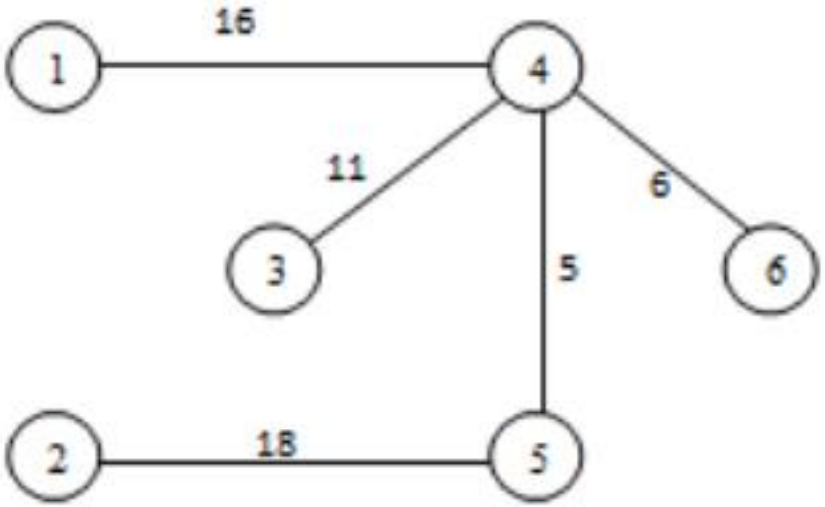
T_{vi} = Violation Time at store i
d_i = Demand at store i
$[a_i, b_i]$ = Time Window of store i
T_{s1} = Service Time at store i
M_{ti} = Type of delivered material
Q = Set of Vehicle Capacities at each depot

3. Minimum Spanning Tree Problem

Tree: A connected graph consisting of ‘n’ nodes and ‘n-1’ arcs without circuits (loops).



Graph



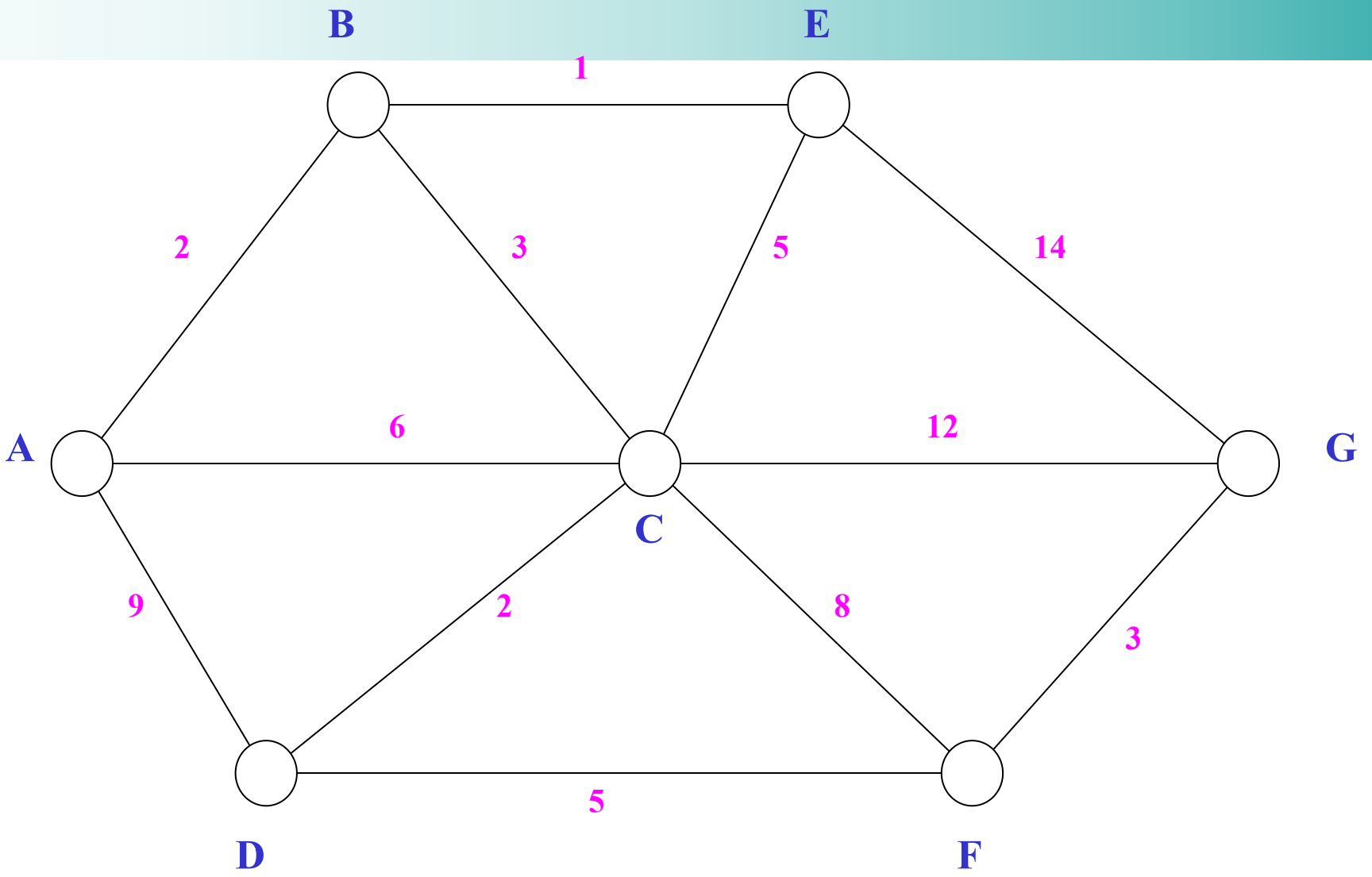
Minimum Spanning Tree

Spanning Tree: A sub-graph of a graph which is a tree and spans all nodes of the graph.

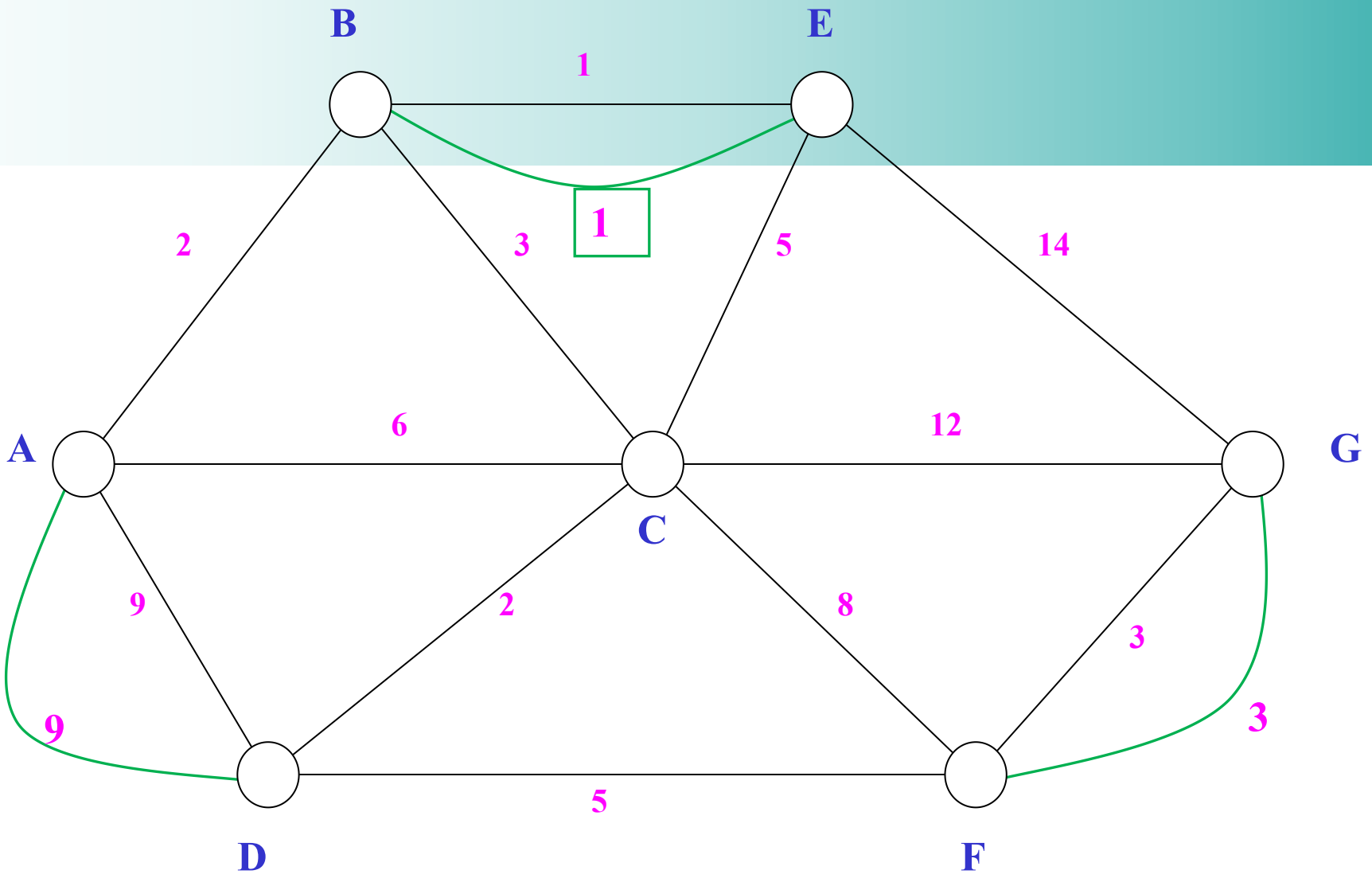
Minimum Spanning Tree: A spanning tree of graph with minimum weight.

Forest: A graph that contains no cycle is a forest.

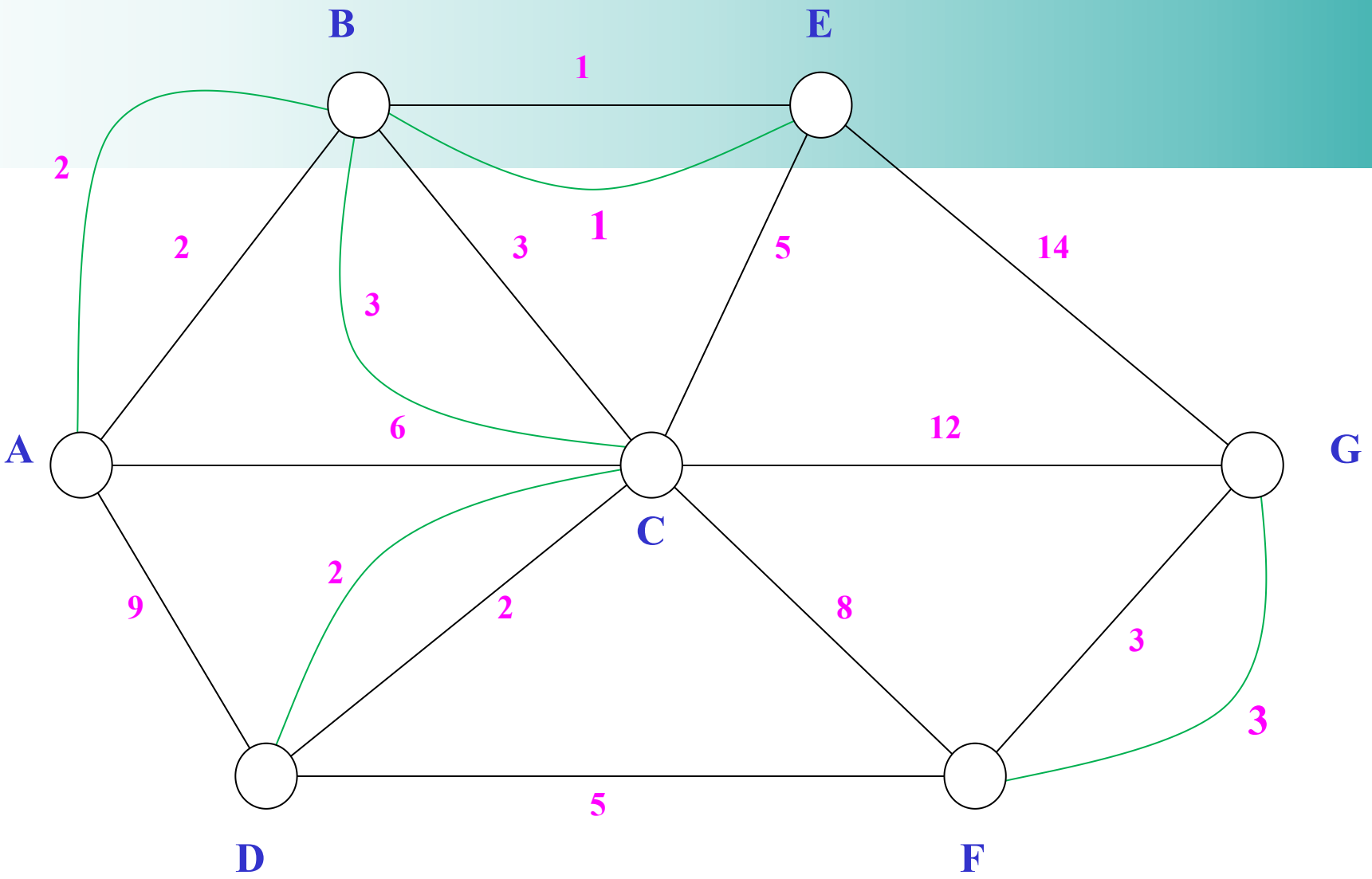
4. Chinese Postman Problem



Eulerian trail starting and ending at A not possible. Why not?



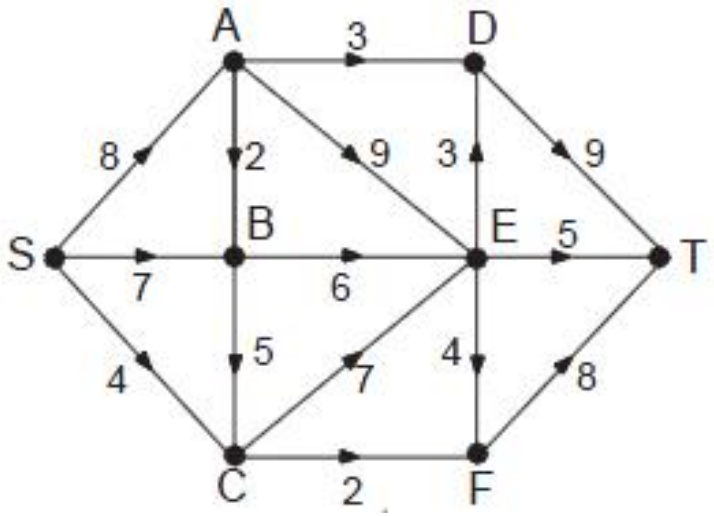
Best Solution?



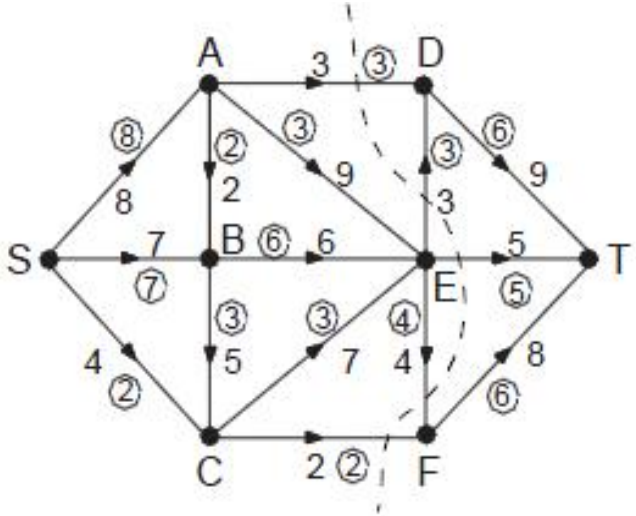
Best solution!

5. Maximum Flow Problem

In a capacitated network, we wish to send as much flow as possible between two special nodes, a source node s and a sink node t, without exceeding the capacity of any arc



Capacitated Network



Augmented Path cum Residual Network

Practical examples

- liquids flowing through pipes
- current through electrical network
- goods transported on the road…
- traffic flow through roads
- information through communication network



THANK YOU

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